Practical No: 5-B

Adapter Based Views

Aim: Develop an Android application using GridView with Custom Adapter and image selection handling.

GridView is a ViewGroup that displays items in a two-dimensional scrollable grid.

A **Custom Adapter** is used to display custom layouts (such as images and text) inside GridView cells.

When a user clicks an image, an event is handled using **OnItemClickListener**.

1) **activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<GridView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/gridView"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity"
  android:numColumns="2"
  android:verticalSpacing="10dp"
  android:horizontalSpacing="10dp"
  android:padding="10dp"/>
```

Step 1

In Android Studio, go to:

```
app
 └── res
     └── layout
```

Step 2

Right-click **layout → New → Layout Resource File**

Step 3

Enter:

File Name : grid_item
Root Element : LinearLayout -> then click on OK

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2) **grid_item.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center">

  <ImageView
    android:id="@+id/imageView"
     android:layout_width="120dp"
     android:layout_height="120dp"
     android:scaleType="centerCrop"/>

  <TextView
     android:id="@+id/textView"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:textSize="18sp"/>

</LinearLayout>
```

3) CustomAdapter.java (Create class file -CustomAdapter.java–Right-click on your package name. Select:New → Java Class. Enter:CustomAdapter Click **OK**.

```
package com.example.linear_lay;
import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.ImageView;
import android.widget.TextView;
public class CustomAdapter extends BaseAdapter {
  Context context;
  String[] names = {
        "Flower",
        "Nature",
        "Mountain",
        "River"
  };
  int[] images = {
        R.drawable.nature,
        R.drawable.mountain,
        R.drawable.river,
```

```
        R.drawable.flower
    };
    public CustomAdapter(Context context) {
        this.context = context;
    }
    public int getCount() {
        return images.length;
    }
    public Object getItem(int position) {
        return images[position];
    }
    public long getItemId(int position) {
        return position;
    }
    public View getView(int position, View convertView, ViewGroup parent) {
        LayoutInflater inflater = LayoutInflater.from(context);
        View view = inflater.inflate(R.layout.grid_item, parent, false);
        ImageView imageView = view.findViewById(R.id.imageView);
        TextView textView = view.findViewById(R.id.textView);
        imageView.setImageResource(images[position]);
        textView.setText(names[position]);
        return view;
    }
}
```

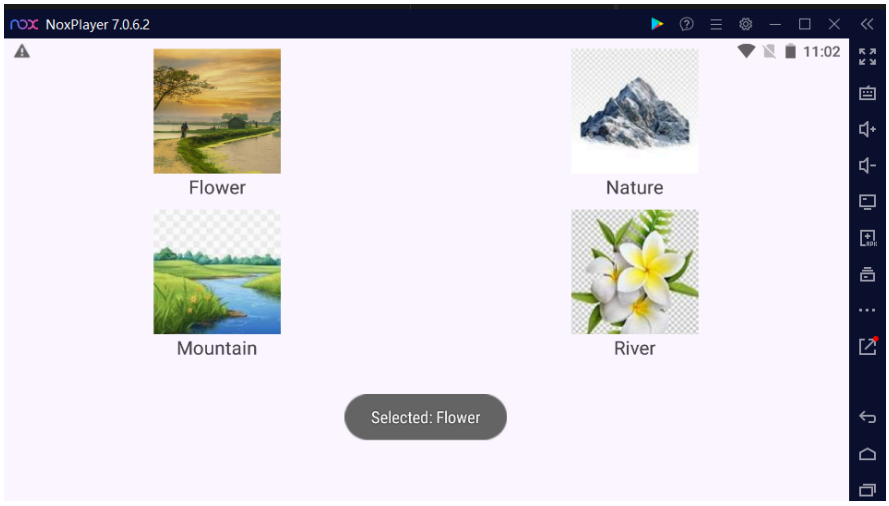
4) MainActivity.java

```
package com.example.linear_lay;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    GridView gridView;
    String[] names = {
```

```
            "Flower",
            "Nature",
            "Mountain",
            "River"
    };
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        gridView = findViewById(R.id.gridView);
        CustomAdapter adapter =
                new CustomAdapter(this);
        gridView.setAdapter(adapter);
        gridView.setOnItemClickListener(
                new AdapterView.OnItemClickListener() {
                    @Override
                    public void onItemClick(
                            AdapterView<?> parent,
                            android.view.View view,
                            int position,
                            long id) {
                        Toast.makeText(MainActivity.this, "Selected: " + names[position],
Toast.LENGTH_SHORT).show();
                    }
                });
    }
}
```

Output:



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